Ridge Regression on Diamonds Dataset

A Comparative Analysis of Optimization Algorithms

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Introduction

The objective of this study is to conduct a comparative analysis of the convergence behavior of various optimization algorithms applied on a Ridge Regression problem to the Diamonds Dataset.

In particular, the focus is on:

- Evaluating the respective algorithms convergence rates.
- Investigating how theoretical properties of the objective function (such as Convexity, Smoothness ecc..) have an impact to results.
- Analyzing how the L_2 regularization parameter (λ) of Ridge Regression influences algorithms behavior.

The following algorithms were implemented:

- Gradient Descent (Naive, Bounded, Smooth).
- Accelerated Gradient Descent (Nesterov's).
- Mini-Batch Stochastic Gradient Descent (Decaying Learning rate)
- Newton's Method.
- Adagrad.

Dataset Overview: The Diamonds Dataset

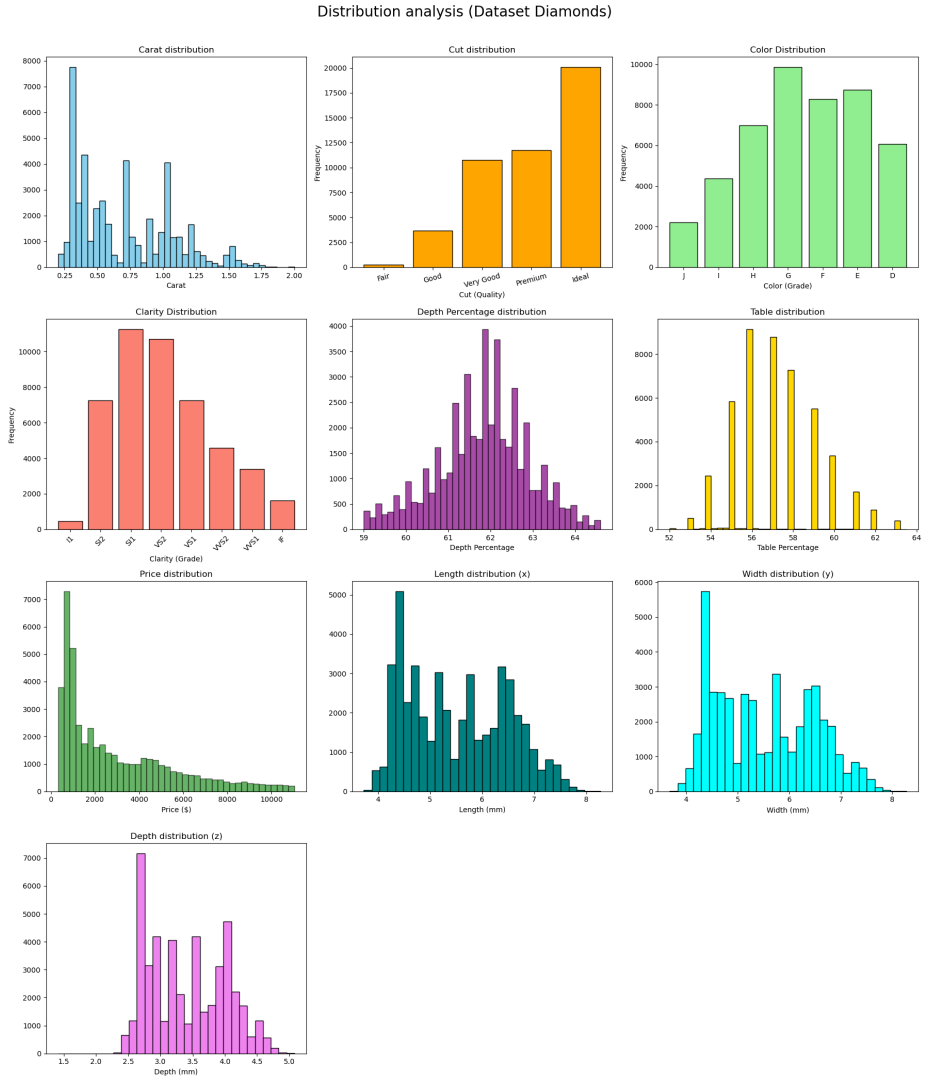
We utilized the **Diamonds Dataset**. The dataset consists of approximately **54,000 observations** ($n = 53.940$), each representing a diamond with **10 distinct attributes**. The goal is to predict with a Ridge Regression the **Price** of a diamond based on its physical and quality characteristics.

Feature Description:

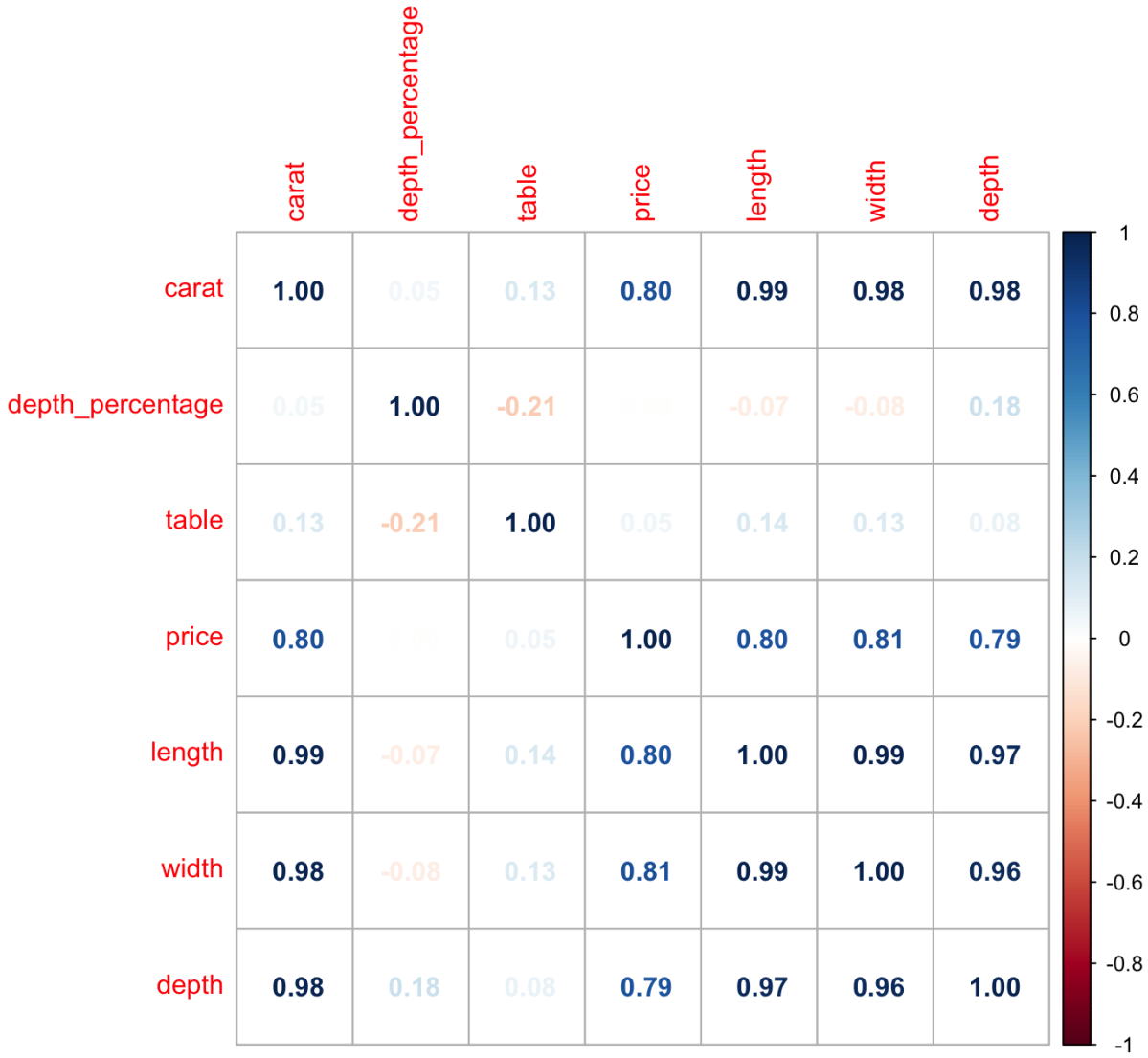
- **Target Variable (y):** `price` in US dollars.
- **Numerical Features:**
  - `carat`: Weight of the diamond.
  - `x`, `y`, `z`: Length, Width, and Depth in mm.
  - `depth`: The depth of the diamond divided by the diameter of the diamond.
  - `table`: The length of the table facet divided by the diameter of the diamond.
- **Categorical Features:**
  - `cut`: Quality of the cut (Fair, Good, Very Good, Premium, Ideal).
  - `color`: Diamond color, from J (coloured, worst) to D (colorless, best).
  - `clarity`: How clear the material is and what surface imperfections it have.

Dataset Overview: The Diamonds Dataset

Distribution of all the main variables within the diamonds dataset.



Dataset Characteristics: Feature matrix



Dataset Characteristics: Feature correlation

Multicollinearity:
In our dataset, there is a strong correlation between all variables, in particular between `carat` (weight) and the physical dimensions (`x`, `y`, `z`).

Impact on Optimization:

- Given the high multicollinearity among the features in the matrix X, we expect small eigenvalues for the Normal Matrix $X^T X$ and a loss objective landscape with flat and elongated valleys.
- Consequently, the optimization problem is **ill-conditioned** and particularly challenging for gradient-based methods.
- We will demonstrate that we can mitigate this issue by using Ridge Regression instead of standard linear regression.

Ridge Regression

Ridge Regression problem extends Ordinary Least Squares (OLS) by imposing an L_2 penalty on the model parameters to mitigate overfitting and multicollinearity.

The objective function is the mean squared errors for each data point ($i = 1$ to n) plus the sum of squared weights (from $j = 1$ to d).

$$f(w) = \frac{1}{2n} \sum_{i=1}^{n} (y_i - x_i^T w)^2 + \frac{\lambda}{2} \sum_{j=1}^{d} w_j^2$$

We can define it in vector notation as follows:

$$f(w) = \underbrace{\frac{1}{2n} \|y - Xw\|^2}_{\text{Half MSE}} + \underbrace{\frac{\lambda}{2} \|w\|^2}_{\text{Regularization}}$$

Where:

- $X \in \mathbb{R}^{n \times d}$ is the feature matrix (includes a column of 1).
- $y \in \mathbb{R}^n$ is the target variable (price).
- $w \in \mathbb{R}^d$ is the weights vector.
- $\lambda > 0$ is the regularization hyperparameter.

Gradient and Hessian Matrix

Since $f(w)$ is differentiable everywhere, we can compute the first-order derivative (the Gradient) and second-order derivative (the Hessian matrix).

We will use them later in optimization's algorithm and to investigate the property of $f(w)$.

The Gradient $\nabla f(w)$

$$\nabla f(w) = \frac{1}{n} X^T (Xw - y) + \lambda w$$

The Hessian Matrix $\nabla^2 f(w)$

$$H = \nabla^2 f(w) = \frac{1}{n} X^T X + \lambda I$$

$f(w)$ is strictly convex

To investigate the convexity of our objective function, we rely on the Second-order Characterization of Convexity:

- *If $\nabla^2 f(\mathbf{x})$ exists at every point $\mathbf{x} \in \mathrm{dom}(f)$ and is symmetric. Then f is convex if and only if $\mathrm{dom}(f)$ is convex, and for all $\mathbf{x} \in \mathrm{dom}(f)$, we have $\nabla^2 f(\mathbf{x}) \succeq 0$. So $\nabla^2 f(\mathbf{x})$ is* **positive semidefinite**.
- *A symmetric matrix M is Positive Semi-Definite if $v^T Mv \geq 0$ for all vectors v, and Positive Definite if $v^T Mv > 0$ for all non-zero vectors $v \neq 0$.*

Let's apply this definition to our Ridge Regression Hessian matrix, analyzing its quadratic form $v^T Hv$ for any arbitrary non-zero vector $v \in \mathbb{R}^d$:

$$v^T Hv = v^T \left(\frac{1}{n} X^T X + \lambda I \right) v$$

$f(w)$ is strictly convex

Solving the quadratic form, we get:

$$v^T H v = \underbrace{\frac{1}{n}\|Xv\|^2}_{\text{OLS Base}} + \underbrace{\lambda\|v\|^2}_{\text{Ridge Penalty}}$$

- **The OLS Base:** The squared norm $\|Xv\|^2$ is a squared length, meaning it is always non-negative (≥ 0).
- **The Ridge Penalty:** For any non-zero vector $v \neq 0$, its squared norm $\|v\|^2$ is strictly positive. Since our regularization hyperparameter is strictly positive ($\lambda > 0$), the entire term $\lambda\|v\|^2$ is strictly greater than zero (> 0).

Because we are adding a strictly positive term to a non-negative term, the entire sum is guaranteed to be strictly positive.

Adding the ridge penalty guarentee us that Hessian matrix H is **Positive Definite** and consequently the objective function possesses a single unique global minimum, so that means $f(w)$ is strictly convex.

Ridge Regression - Properties

To understand how optimization algorithms will behave on Ridge Regression objective function $f(w)$, we must analyze its properties.

1. Convexity: Yes (Strictly Convex)
As proven in the previous section, the addition of the L_2 penalty ensures that the Hessian matrix H is Positive Definite. Therefore, the function is not just convex, but **strictly convex**. This guarantees that any local minimum is the unique global minimum w^*.

2. Lipschitz Continuous: No (Globally)
A function $f(w)$ is globally Lipschitz continuous if its gradients are bounded in norm everywhere ($\|\nabla f(w)\| \leq B$). Our objective function $f(w)$ is a quadratic function. As the weights w can grow toward infinity, the gradient $\nabla f(w) = \frac{1}{n} X^T(Xw - y) + \lambda w$ also can grows linearly toward infinity. Because the gradient is unbounded over the entire domain $\mathbb{R}^d$, the function itself is **not globally Lipschitz continuous**.

Ridge Regression - Properties

3. Smoothness (L-smooth): Yes
It is L-**Smooth**. The curvature does not change abruptly and the gradient doesn't oscillate wildy, the function always lies *below* a certain parabola:

$$f(y) \leq f(x) + \nabla f(x)^T (y - x) + \frac{L}{2}\|x - y\|^2 \quad \forall x, y$$

- **Value of L:** It corresponds to the largest eigenvalue of the Hessian matrix (the maximum curvature of the loss landscape).

$$L = \lambda_{\max}\left(\frac{1}{n} X^T X\right) + \lambda$$

- **Consequence:** This value acts as the absolute **speed limit** for Gradient Descent. The optimal safe step size is $\gamma = \frac{1}{L}$.

Ridge Regression - Properties

4. Strong Convexity (μ-strongly convex): Yes
It is μ-**Strongly Convex**. The curvature is "at least" μ in every direction. The function always lies *above* the quadratic function:

$$f(y) \geq f(x) + \nabla f(x)^T (y - x) + \frac{\mu}{2} \|x - y\|^2 \quad \forall x, y$$

- **Value of μ:** It corresponds to the smallest eigenvalue of the Hessian matrix.

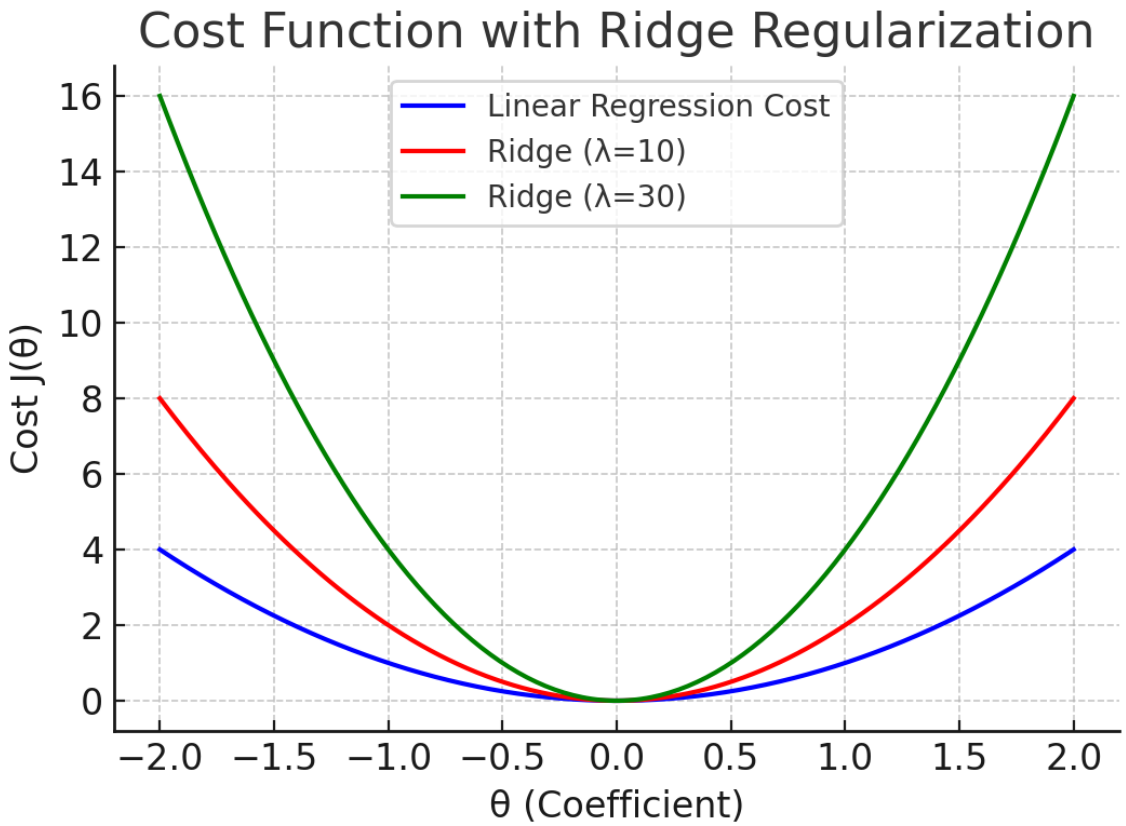
$$\mu = \lambda_{\min} \left(\frac{1}{n} X^T X \right) + \lambda$$

- **Consequence:** μ indicates how much the function is strongly convex.

The effect of $L2$ regularization

Specifically, increasing λ directly increases μ, the **strong convexity parameter**.

When we set a large λ, the minimum curvature (μ) is strictly bounded away from zero, the loss function becomes steeper and we eliminate flat regions where Gradient's based method might stall.



Dataset pre-processing steps

To prepare the dataset for our optimization algorithms, we now apply the following preprocessing steps:

- **Ordinal Encoding:** Categorical variables (`cut`, `color`, `clarity`) were mapped to integer values, preserving their inherent hierarchy.
- **Removing Outliers:** We filter out extreme anomalous data points.
- **Standardize the Features (X) and the Target (y):** By subtracting the mean and dividing by the standard deviation, we ensure all features and the target share the same scale (a mean of 0 and a variance of 1), also it allows Gradient Descent to point much more directly toward the minimum.
- **The Bias Trick (Adding a column of 1s):** We append a column of ones to our feature matrix X to absorb the intercept (or bias) term directly into our weight vector w.

Testing approach

To test the performance of the optimization algorithms, we use the following framework:

1. The Methodology:

- We create a predefined set of λ (regularization) values: $\lambda \in \{10^{-5}, 10^{-4}, 10^{-3}, 0.01, 0.1, 1.0\}$.
- We define five Ridge Regression problems, one fore each specific λ in the set, and we will apply every optimization algorithm to them.

2. The Goals:

By establishing these different regularization classes, we can evaluate:

- **Regularization Impact:** How varying regularization conditions affects the convergence speed and performance of the *same* algorithm.
- **Inter-Algorithm Comparison:** How *different* algorithms perform against each other when subjected to the *same* regularization conditions.

Testing approach

3. Early Stopping Mechanism:

- At each iteration, we evaluate if the algorithm has successfully reached the minimum of the loss function:

$$\epsilon = f(w_t) - f(w^*) \leq 10^{-6}$$

- If convergence is achieved, execution terminates immediately, returning the elapsed time and total iterations.
- Otherwise, the algorithm continues until it hits a predefined `max_iters` limit.

4. Exact Analytic Solution (Benchmark):

- We have to compute the exact optimal solution for every Ridge Regression subproblem to use it as a benchmark.
- We use the closed-form solution by `np.linalg.solve(A, b)`.

λ	10^{-5}	10^{-4}	10^{-3}	0.01	0.1	1.0
$f(w^*)$	0.04265771	0.04276069	0.04369175	0.04977553	0.07101217	0.16355421

Gradient Descent - Naive Learning Rate

Gradient descent by definition updates the weight vector w step-by-step by moving in the opposite direction of the gradient, scaled by the learning rate (step size) γ:

$$w^{(t+1)} = w^{(t)} - \gamma \nabla f(w^{(t)})$$

For this first run, we are using a **naive, very small learning rate $\gamma = 0.0001$** .

Choosing a very small step size is a conservative and highly stable approach: it guarantees that our algorithm will not overshoot the minimum or diverge.

But this safety comes with a heavy trade-off in efficiency: taking such microscopic steps means the algorithm might require a massive number of iterations to finally reach the optimal solution.

Gradient Descent - Naive Learning Rate

Final results per λ using a naive learning rate ($\gamma = 0.0001$):

λ	**Loss**	γ	**Iters**	**Time[s]**
10^{-5}	0.055102	0.0001	50000	9.55
10^{-4}	0.055118	0.0001	50000	10.53
10^{-3}	0.055281	0.0001	50000	9.70
0.01	0.056890	0.0001	50000	10.24
0.1	0.071706	0.0001	50000	11.27
1.0	0.163555	0.0001	31209	6.40

Observations:

- For low regularizations (from $\lambda = 10^{-5}$ to 0.1), the algorithm hits the `max_iters` limit of 50.000 without reaching the tolerance of the early stop.
- Only for the strongest regularization ($\lambda = 1.0$), the algorithm is pushed fast enough to converge in exactly 31.209 iterations.

Gradient Descent - Assuming Bounded Gradient

Assume the distance between initial solution and optimal solution is bounded by R and this bounded region contains all iterates.

Since we initialize our initial solution vector to **Zero** (`np.zeros`), the distance coincides exactly with the norm of the solution:

$$\|w_0 - w^*\| \leq R$$

$$\|0 - w^*\| = \|w^*\| \leq R$$

The assumption on R means: *"The optimal solution and the optimization path to reach it lie within a sphere of radius R."*

- Since we have **standardized the data** (X and y have zero mean and unit variance), the optimized weights w will naturally be small (typically between -1 and 1)
- We set $R = 5.0$. A generous safety margin (a "Bounded Region") that is virtually guaranteed to contain the true solution.

Gradient Descent - Assuming Bounded Gradient

We need to find a number B such that $\|\nabla f(w)\| \leq B$ for every w within the radius R. We rewrite the gradient formula in this way:

$$\nabla f(w) = \left(\frac{1}{n} X^T X + \lambda I\right) w - \frac{1}{n} X^T y$$

Using the triangle inequality ($\|a + (-b)\| \leq \|a\| + \|b\|$) and the definition of matrix norms:

$$\|\nabla f(w)\| \leq \underbrace{\left\|\frac{1}{n} X^T X + \lambda I\right\|}_{\text{This is smooth parameter L}} \cdot \underbrace{\|w\|}_{\leq R} + \underbrace{\left\|\frac{1}{n} X^T y\right\|}_{\text{Constant}}$$

So the formula for B is:

$$B = L \cdot R + \left\|\frac{1}{n} X^T y\right\|$$

Gradient Descent - Assuming Bounded Gradient

By making the assumption that our optimization path remains within a bounded region $\|w_0 - w^*\| \leq R$, and having calculated the maximum gradient bound B such that $\|\nabla f(w)\| \leq B$ for all w, we have effectively established that our objective function is **Lipschitz continuous** within this domain.

For Lipschitz continuous functions, optimization theory provides a specific, optimal step size: we can now set γ using the formula:

$$\gamma = \frac{R}{B\sqrt{T}}$$

where $T = 10.000$ iterations.

Gradient Descent - Assuming Bounded Gradient

Final results per λ using the bounded gradient step size (γ):

λ	**Loss**	γ	**Iters**	**Time[s]**
10^{-5}	0.049663	0.00214	10000	1.86
10^{-4}	0.049692	0.00214	10000	2.04
10^{-3}	0.049974	0.00214	10000	2.88
0.01	0.052629	0.00214	10000	2.15
0.1	0.071028	0.00210	10000	2.02
1.0	0.163555	0.00176	1768	0.32

Observations:

- **Larger Steps:** The dynamic $\gamma \approx 0.002$ is safely 20x larger than the naive guess.
- **Massive Speedup:** For $\lambda = 1.0$, it converges in only 1,768 iterations (vs. 31,209).
- **Higher Accuracy:** It reaches a strictly lower loss in just 10,000 steps compared to the 50,000 steps of the naive approach.

Gradient Descent - Smoothness

Previously we have seen that function is L-**Smooth**.

- **Value of L:** It is the largest eigenvalue of the Hessian matrix (the maximum curvature).

$$L = \lambda_{\max}\left(\frac{1}{n}X^T X\right) + \lambda$$

or equivalently

$$L = \left\|\frac{1}{n}X^T X\right\| + \lambda$$

- **Consequence:** This value is the **speed limit** for Gradient Descent.The optimal step size is $1/L$.

Gradient Descent - Smoothness

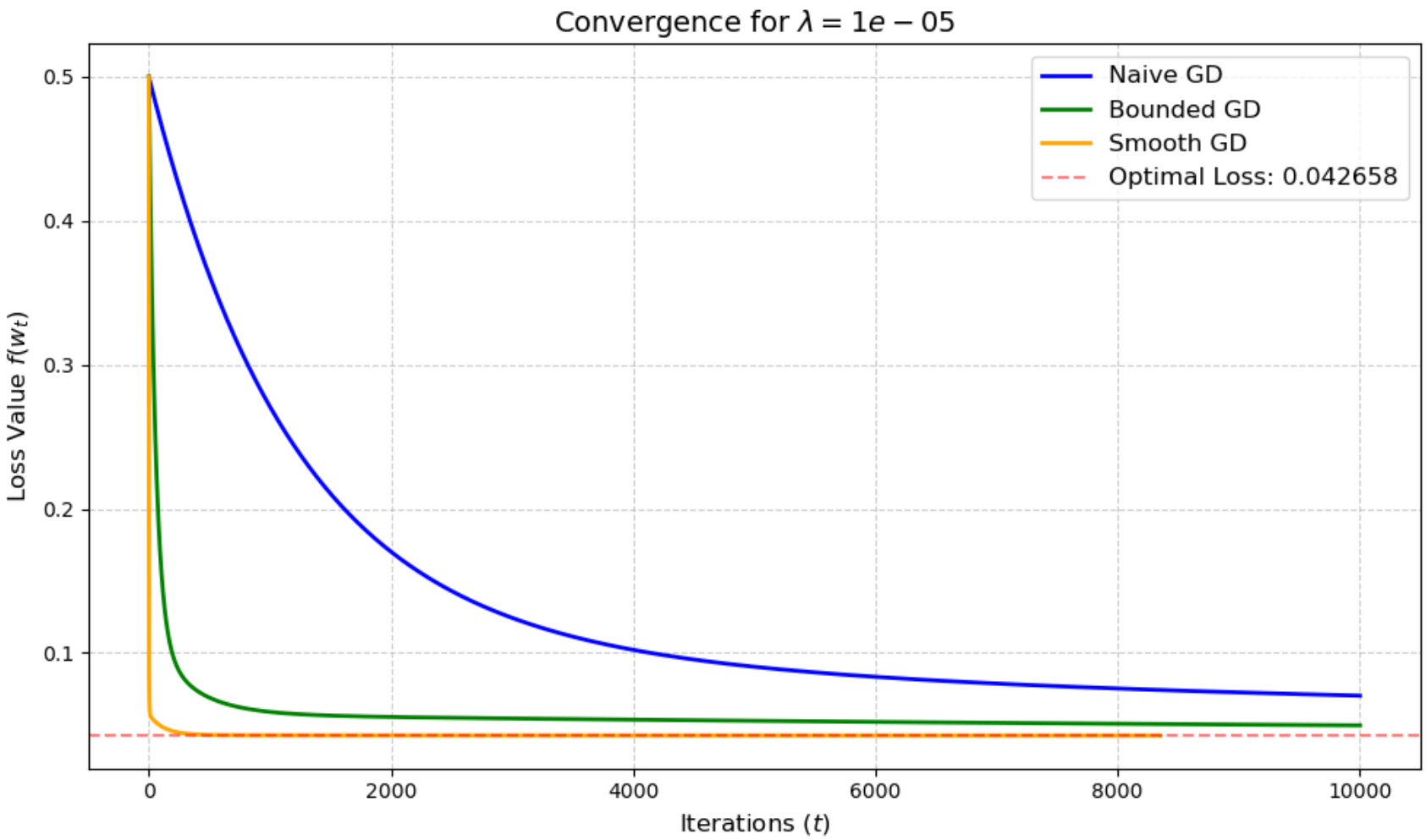
Final results per λ using the optimal step size ($\gamma = 1/L$):

λ	**Loss**	$\gamma = 1/L$	**Iters**	**Time[s]**
10^{-5}	0.042659	0.23247	8353	1.56
10^{-4}	0.042762	0.23247	7767	1.74
10^{-3}	0.043693	0.23242	4438	1.76
0.01	0.049777	0.23193	765	0.22
0.1	0.071013	0.22719	142	0.07
1.0	0.163555	0.18862	15	0.01

Observations:

- **Optimal Steps:** The safe learning rate ($\gamma \approx 0.23$) is $\sim$2,300x larger than our naive guess.
- **Lightning Fast:** For $\lambda = 1.0$, it perfectly converges in just 15 iterations (down from 31.209).
- **Perfect convergence:** For all case the algorithm converge to optimal solution.

Convergence Analysis: Comparing Gradient Descent implementations



Nesterov's Accelerated Gradient Descent

Now we implement the **Accelerated Gradient Descent** algorithm. This method improves standard gradient descent by computing the next iteration point as a dynamically weighted average between a conservative "smooth" step and a more "aggressive" step.

Choosing arbitrary initial points $\mathbf{z}_0 = \mathbf{u}_0 = \mathbf{w}_0$, for $t \geq 0$ we set:

$$\mathbf{u}_{t+1} := \mathbf{w}_t - \frac{1}{L}\nabla f(\mathbf{w}_t)$$

$$\mathbf{z}_{t+1} := \mathbf{z}_t - \frac{t+1}{2L}\nabla f(\mathbf{w}_t)$$

$$\mathbf{w}_{t+1} := \frac{t+1}{t+3}\mathbf{u}_{t+1} + \frac{2}{t+3}\mathbf{z}_{t+1}$$

- `w` represents the main weight vector.
- `u` represents the prudent/smooth update.
- `z` represents the aggressive update.

Nesterov's Accelerated Gradient Descent

Final results per λ using Accelerated Gradient Descent:

λ	**Loss**	**Iters**	**Time[s]**
10^{-5}	0.042659	203	0.10
10^{-4}	0.042761	202	0.12
10^{-3}	0.043692	152	0.07
0.01	0.049776	86	0.05
0.1	0.071012	23	0.01
1.0	0.163555	12	0.005

Observations:

- **Massive Speedup:** For ill-conditioned problem ($\lambda = 10^{-5}$), iterations go down from 8,353 to just 203 ($\sim$40x improvement).
- **Faster Convergence Rate:** Validates the theoretical improvement from $O(1/\epsilon)$ to $O(1/\sqrt{\epsilon})$.
- **Diminishing Returns:** For $\lambda = 1.0$ (a steep, well-conditioned bowl), the advantage is marginal (12 vs. 15 iterations).

Stochastic Gradient Descent with Mini-batches

Mini-batch Stochastic Gradient Descent (SGD) approximate the true gradient using a much smaller, randomly selected subset of the data of size m (where $m < n$).

Starting from an initial point $\mathbf{x}_0 \in \mathbb{R}^d$, at each iteration t we choose a random subset of indices $\{h_1, \ldots, h_m\} \subseteq \{1, \ldots, N\}$ and update the weights using a learning rate $\gamma_t \geq 0$:

$$\mathbf{w}_{t+1} := \mathbf{w}_t - \gamma_t \frac{1}{m} \sum_{j=1}^{m} \nabla f_{h_j}(\mathbf{w}_t)$$

- The algorithm operates in "epochs". At the start of an epoch, the entire dataset is randomly shuffled (`np.random.permutation`).
- Within an epoch, a loop sequentially iterates over this shuffled data and cut them in chunks of size `batch_size` $= 1000$ (our m).
- At each iteration the `minibatch_gradient` function calculates gradient on a small chunk, and the main loop updates the weights.

Mini batches SGD - Decaying Learning Rate

Since our objective function is μ-strongly convex. From **Tame Strong Convexity**, the learning rate to use is:

$$\gamma_t = \frac{2}{\mu(t+1)}$$

and it decays over each iteration.

However, applying this formula directly when our regularization parameter λ is small leads to immediate divergence.

- Due to multicollinearity of the dataset, the Hessian matrix of our objective function has very small eigenvalues.
- If μ is very small, the initial step size γ_t becomes too large, causing the algorithm to diverge.

Mini batches SGD - Decaying Learning Rate

We can demonstrate it: for our smallest ridge penalization ($\lambda = 10^{-5}$), the minimum curvature of our loss function is extremely flat, resulting in $\mu \approx 0.00130$. If we plug this value into the pure **Tame Strong Convexity** formula to calculate the learning rate for our very first iteration ($t = 0$), we get:

$$\gamma_0 = \frac{2}{0.00130} \approx 1538$$

This initial step size is catastrophically large and massively violates the theoretical speed limit $(2/L) \approx 0.46$ that prevents divergence for L-Smooth functions.

This same issue occurs across all other small values of the ridge regularization parameter λ.

Mini batches SGD - Decaying Learning Rate

To safely apply the optimal decay rate by the **Tame Strong Convexity**, we propose an adjusted learning rate:

$$\gamma_t = \frac{2}{L + \mu(t+1)}$$

In this way we get:

- **Initial Stability ($t = 0$):** At the very first iteration, the learning rate evaluates to exactly $\frac{2}{L+\mu} < \frac{2}{L}$. This respect the theoretical speed limit for smooth strongly convex functions and prevents the algorithm from diverging, regardless of how small μ is.

- **Asymptotic Convergence ($t \to \infty$):** As the iterations progress, the L term becomes dominated by μt. The learning rate transitions to approximate $\frac{2}{\mu t}$, like in the original Tame Strong Convexity formula.

Mini batches SGD - Decaying Learning Rate

Final results per λ using the adjusted decaying learning rate (γ_t):

λ	**Loss**	**Iters**	**Time[s]**
10^{-5}	0.042659	14199	3.58
10^{-4}	0.042761	15115	3.94
10^{-3}	0.043693	8104	2.23
0.01	0.049776	1730	0.71
0.1	0.071013	404	0.13
1.0	0.163555	101	0.04

Observations:

- **Guaranteed Convergence:** Safely bounded initial steps eliminate chaotic bouncing, achieving perfect convergence to the exact global minimum for every λ, but the stochastic noise makes Mini-Batch SGD slower compared to Smooth-GD.

Newton's Method

Until now, we have relied on first-order optimization algorithms (like Gradient Descent and its variants), which only use the gradient (the slope) to update weights.

Newton's Method introduces a massive upgrade by incorporating second-order derivative information—specifically, the curvature of the loss landscape represented by the Hessian matrix $\nabla^2 f(\mathbf{x})$.

Mathematically, for minimizing a convex function $f : \mathbb{R}^d \to \mathbb{R}$ in the d-dimensional case, the update rule is defined as:

$$\mathbf{w}_{t+1} := \mathbf{w}_t - \nabla^2 f(\mathbf{w}_t)^{-1} \nabla f(\mathbf{w}_t)$$

Newton's Method

Final results per λ using Newton's Method:

λ	**Loss**	**Iters**	**Time[s]**
10^{-5}	0.042658	1	0.006
10^{-4}	0.042761	1	0.004
10^{-3}	0.043692	1	0.003
0.01	0.049776	1	0.003
0.1	0.071012	1	0.003
1.0	0.163554	1	0.002

Observations:

- **Single update iteration:** Newton's Method achieves exact numerical convergence (to our 10^{-6} tolerance) in a single update step.

Newton's Method

This result is a direct consequence of the underlying mathematical nature of **Ridge Regression**. The objective function is a quadratic and strictly convex function with respect to the weight vector w.

$$f(w) = \frac{1}{2n}\|Xw - y\|^2 + \frac{\lambda}{2}\|w\|^2$$

The One-Step Convergence Lemma:
On nondegenerate quadratic functions, with any arbitrary starting point $w_0 \in \mathbb{R}^d$, Newton's method yields the global minimum w^* in exactly one step ($w_1 = w^*$).

Because the Ridge Regression objective function satisfies the condition of a nondegenerate quadratic function (a matrix with strictly positive eigenvalues has a non-zero determinant and is therefore invertible), this experimental behavior aligns exactly with optimization theory.

Adagrad

Adagrad modifies standard Stochastic Gradient Descent by introducing a dynamic, per-coordinate learning rate. At each iteration t, we first pick a stochastic gradient $\mathbf{g}_t$.

1. Gradient Accumulation
We accumulate the history of squared gradients for each individual coordinate i:

$$[G_t]_i := \sum_{s=0}^{t} ([\mathbf{g}_s]_i)^2 \quad \forall i$$

(This step tracks how strong the gradients have been in the past for the i-th coordinate).

2. Parameter Update
We then update each coordinate i of our parameter vector $\mathbf{w}$:

$$[\mathbf{w}_{t+1}]_i := [\mathbf{w}_t]_i - \frac{\gamma}{\sqrt{[G_t]_i}} [\mathbf{g}_t]_i \quad \forall i$$

Adagrad

The Adaptive Learning Rate:
The crucial part of this formula is the term $\frac{\gamma}{\sqrt{[G_t]_i}}$, which acts as our new coordinate-specific learning rate.

- By dividing the base learning rate γ by the square root of the accumulated gradients, the algorithm automatically **reduces the step size** for coordinates that have historically had large gradients.
- Instead, if a coordinate has had consistently small gradients, its learning rate remains relatively **higher**.

This mechanism speeds up convergence and reduces oscillations along directions where gradients are already large.

Adagrad

Final results per λ using the Adagrad algorithm with $\gamma = 0.1$:

λ	**Loss**	**Iters**	**Time[s]**
10^{-5}	0.042659	42511	11.99
10^{-4}	0.042762	37110	9.61
10^{-3}	0.043693	19376	4.73
0.01	0.049776	3477	0.78
0.1	0.071013	9059	1.98
1.0	0.163554	8292	1.92

Observations:

- **Guaranteed Convergence:** Adagrad successfully reached the exact global minimum for all λ values.
- **Strong γ Decay:** The accumulation term G_t drives the learning rate to zero very quickly, increasing the number of iterations required to achieve convergence.

Summary of Results for $\lambda = 10^{-5}$

Performance comparison on the ill-conditioned problem ($\lambda = 10^{-5}$):

Optimization Method	**Loss**	**Iterations**	**Time[s]**
Naive Gradient Descent	0.055102	50000*	∼10.50
Bounded Gradient Descent	0.049663	10000*	1.86
Smooth Gradient Descent	**0.042659**	8353	1.56
Accelerated Gradient Descent	**0.042659**	203	0.10
Mini-Batch SGD	**0.042659**	14199	3.58
Newton's Method	**0.042658**	1	0.006
Adagrad	**0.042659**	42511	11.99

**Hit maximum iterations limit without converging*

Summary of Results for $\lambda = 1.0$

Performance comparison on the strongly regularized problem ($\lambda = 1.0$):

Optimization Method	**Loss**	**Iterations**	**Time[s]**
Naive Gradient Descent	**0.163555**	31209	6.39
Bounded Gradient Descent	**0.163555**	1768	0.32
Smooth Gradient Descent	**0.163555**	15	0.01
Accelerated Gradient Descent	**0.163555**	12	0.005
Mini-Batch SGD	**0.163555**	101	0.04
Newton's Method	**0.163554**	1	0.002
Adagrad	**0.163554**	8292	1.92

Conclusions

- **Impact of λ on Convergence:** Increasing the regularization hyperparameter λ makes the objective function more strongly convex and the optimization's problem becomes better conditioned. Helps algorithms to converge and drastically reduces iterations and execution time.

- **Algorithm Performance Comparison:**
  - **Newton's Method:** Converges in exactly 1 step across all λ by leveraging second-order curvature.
  - **Accelerated GD:** Best first-order method, can effortlessly navigates flat valleys.
  - **Smooth GD ($1/L$):** Highly efficient for well-conditioned (high-λ) problems, but struggles heavily in flat scenarios.
  - **Mini-batch SGD:** Safely converges to the minimum, but its noisy updates make it less efficient than the deterministic Smooth-GD.
  - **Adagrad:** Auto-tunes step sizes without theoretical parameters, but the monotonically shrinking learning rate causes slow convergence.